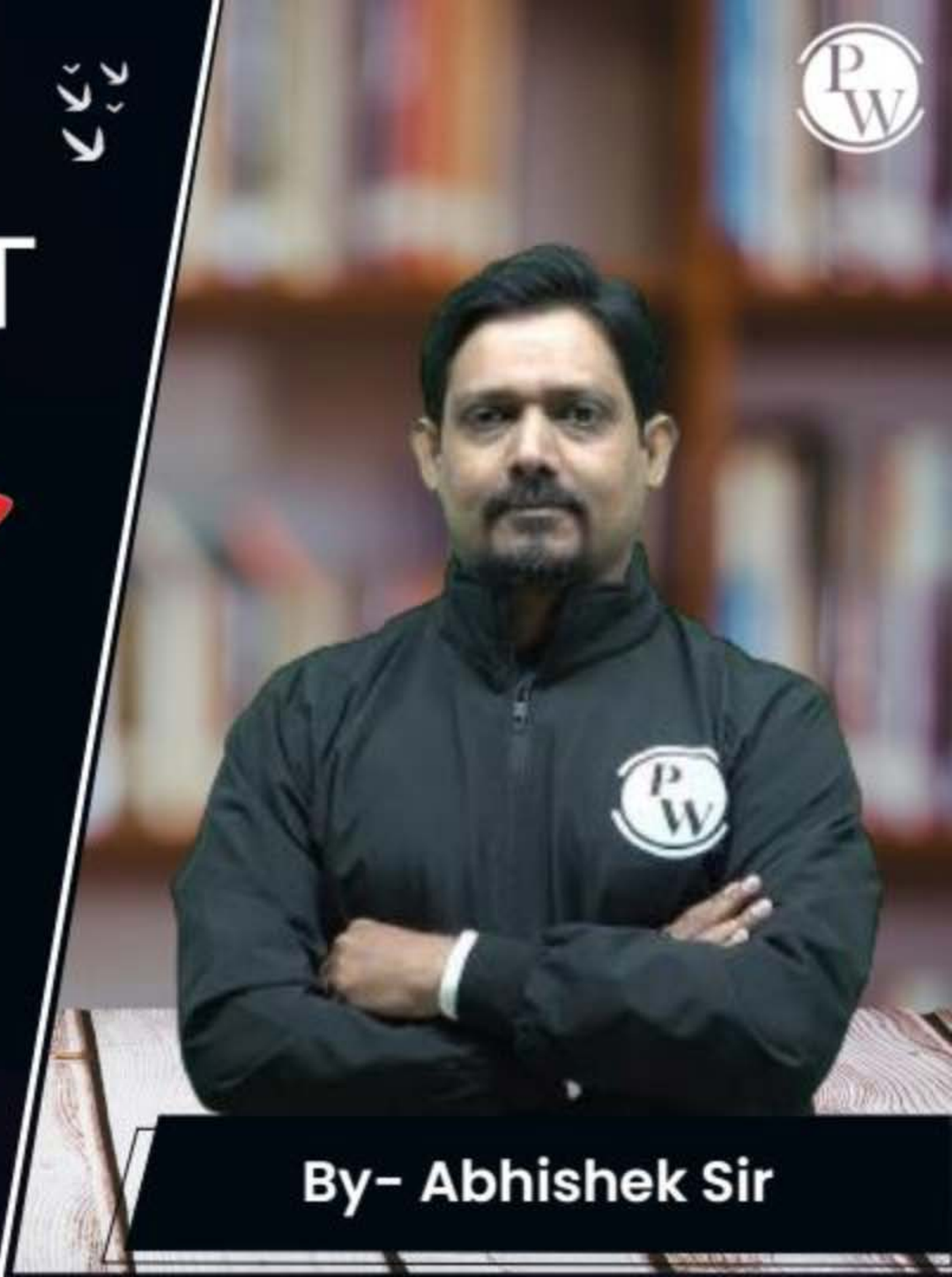
Computer Science & IT



Programming in C

Array & Pointers

Lecture No. 05



By- Abhishek Sir

Recap of Previous Lecture



10 am

Topic — 1 D data type

Topic — 2D array

Topic — Access 2-D array

Topic — A[i][j] = *(*(A+i)+j)

Topic

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Practice Question

2-D array declaration

} String, array of character

Slide

2-D Array

```
#include <stdio.h>

int main(){
    int a[2][2] = {{11,12},{13,14}};

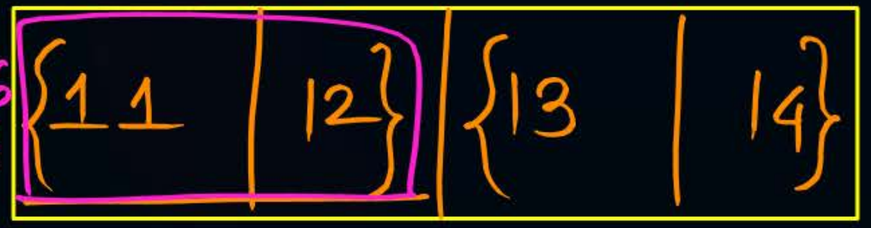
    printf("\n %u",a);
    printf("\n %u",&a);
    printf("\n %u",*a);
    printf("\n %u",**a);

    printf("\n %u",a+1);
    printf("\n %u",*a+1);
    printf("\n %u",**a+1);
    printf("\n %u",&a+1);

    return 0;
}
```

2D Array Size = 4×4 = 16 B

1-D Array Size : 2×4 = 8B



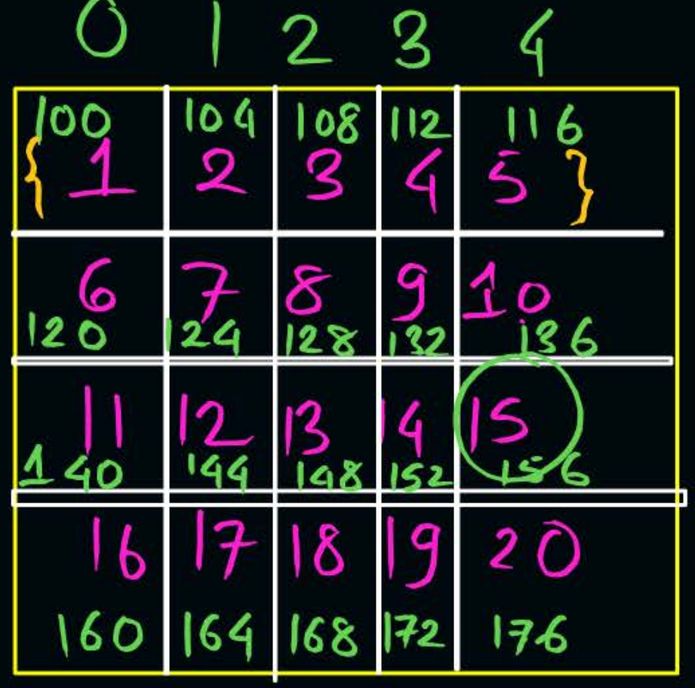
a is Address . 1-D array

*a - Address . Integer

**a = a[0][0] - integer

&a Address of 2-D array

Slide



$5 \times 4 = 20B$

$*(*(a+**a+1)+4)$

$*(*(a+2)+4) = a[2][4] = 15$

$*(*(100+2)+4) = *(*(100+2\times20)+4)$

$*(*(140)+4)$

$*(140+4)$

$*(140+4\times4) = *(156) = 15$

Question

#Q What is the output of the following program ?

```
#include<stdio.h>
int main(){
int i , b [] = {2, 3, 4, 5, 6}, *p ;
p = b ;
++*p ;
printf ("%d\t" ,*p) ;
p += 2 ;
printf ("%d" , *p);
}
```

(A) 1 3
(B) 2 3
(C) 2 4
(D) 3, 4



modify

p 100

3

4

p+=2; = p: p+2 = 100+2 : 100+2×4 = 108

Slide



Question

PYQ 2019

Consider the following C program

```
#include <stdio.h>
int main(){
   int a[] = {2, 4, 6, 8, 10};
   int i, sum = 0, *b = a + 4;
   for (i = 0; i < 5; i++ )
      sum = sum + (*b - i) - *(b - i);
   printf("%d\n",sum);
   return 0;
}
```

The output of the above C-program is _____.



Question

PYQ - 2024

What is the output of the following C program?

```
#include <stdio.h>
int main() {
        double a[2]={20.0, 25.0}, *p, *q;
        p = a;
        q = p + 1;
        printf("%d,%d", (int)(q - p), (int)(*q - *p));
        return 0;
}
```

(A) 4,8
(B) 1,5
(C) 8,5
(D) 1,8

100 108

1

(25.0 - 20.0) → 5.0

double - 8Byte

typecast

p = 100 = 100

q = 100 + 1 = 100 + 1 × 8 = 108

(108 - 100) = 108 - 100 / 8 = 1

Slide

GATE 2015

diagram

Addrss- 8B




```
#include<stdio.h>
int main( ){
      static int a[] = {14,27,73,40,50};
      static int *p[] = {a, a+3,a+4,a+1,a+2};
      int**ptr=p;
      ptr++ ;
      printf ("%d%d", ptr-p, **ptr);
}
```

208-200

The output of the program is 140.

208-200 : (208-200)/8 = 1

208-200 = (208-200)/8 = 8/8 = 1

Slide

declaration

```
int a[2][2];   ✓ correct
int a[][2];    ✗
int a[2][];    ✗
int a[][];     ✗
```

In Initialization you can skip

only first dimension in multidimension array

Access elements will be discarded

Lesser elements will be treated as 0

Question

```
#include<stdio.h>
int main(){
   int a[][2]={1, 3, 5, 7, 9, 11};
   int *ptr=a[1];
   ++*ptr++;
   printf("%d", *ptr);
   return 0;
}
```

The output is-

(a) 5

(b) 9

(c) 7

(d) Compilation error

1-D array Size : 2×4B = 8Byte

1	3	6 ~~5~~	7	9	11
100	104	108	112	116	120

ptr = *(a+1)

= *(100+1) = *(108) = 108

a - Add 1-D

*a Add int.

**a integer

ptr 108 → 112

++*ptr++

++*(108)

2 Increment

112

Question

```
#include<stdio.h>
int main(){
   int a[][2]={1, 3, 5, 7, 9, 11};
   int *ptr=a[1];
   ++*ptr++;
   printf("%d", *ptr);
   return 0;
}
```

The output is-

(a) 5

(b) 9

(c) 7

(d) Compilation error

1-D array Size = 2×4B = 8Byte

1	3	6 ~~5~~	7	9	11
100	104	108	112	116	120

a - 1-D

int *ptr=a[1]; = *(a+1)

= *(100+1) = *(108)

= 108 ← Integer pointer

*a

++*ptr++ → ++*ptr

++*108

ptr | 108 | 112 |

p[] p() p++ ← higher

p→ p.

post higher

++p, &p, *p prefix

-p, ~p, !p



Question

Consider the following ANSI C program

```
#include <stdio.h>
int main() {
  int arr[4] [5]
  int i, j;
  for (i = 0; i< 4; i ++){
       for (j = 0; j < 5; j ++){
            arr[i] [j] = 10*i+j;
       }
  }
  print f ("%d", *(arr[1] + 9));
  return 0;
}
```

What is the output of the above program?

(A) 24
(B) 30
(C) 20
(D) 14

Question

Consider the following ANSI C program

```
#include <stdio.h>
int main() {
  int arr[4] [5]
  int i, j;
  for (i = 0; i< 4; i ++){
       for (j = 0; j < 5; j ++){
            arr[i] [j] = 10*i+j;
       }
  }
  print f ("%d", *(arr[1] + 9));
  return 0;
}
```

arr(1)(9)

	0	1	2	3	4
→ 0	100 0	104 1	108 2	112 3	116 4
→ 1	120 10	124 11	128 12	132 13	136 14
→ 2	140 20	144 21	148 22	152 23	156 24
→ 3	160 30	164 31	168 32	172 33	176 34

((arr+1)+9)

((100+1)+9)

((120)+9)

*(120+9)

=*(120+9×4)

=*(156)

=24



Question

What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[]){
   int x = 1, z[2] = {10, 11};
   int *p = NULL;
   p = &x;
   *p = 10;
   p = &z[1];
   *(&z[0] + 1) += 3;
   printf("%d, %d, %d\n", x, z[0], z[1]);
   return 0;
}
```

(A) 1, 10, 11

(B) 1, 10, 14

(C) 10, 14, 11

(D) 10, 10, 14



Question

What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[])
{
int a[3][3][3] = {{1, 2, 3, 4, 5, 6, 7, 8, 9},
                  {10, 11, 12, 13, 14, 15, 16, 17, 18},
                  19, 20, 21, 22, 23, 24, 25, 26, 27}};
int i = 0, j = 0, k = 0;
for( i = 0; i < 3; i++ ){
     for(k = 0; k < 3; k++ )
          printf("%d ", a[i][j][k]);
     printf("\n");
}
return 0;
}
```

(A) 1 2 3
10 11 12
19 20 21

(B) 1 4 7
10 13 16
19 22 25

(C) 1 2 3
4 5 6
7 8 9

(D) 1 2 3
13 14 15
25 26 27

Slide



2 mins Summary

Topic

Topic

Topic

Topic

Topic

practice problem

2-D declaration

H.W question

Slide


String
THANK - YOU